Notes: Farhi and Tirole (RES, 2021)

Shadow Banking and the Four Pillars of Traditional Financial Intermediation

Contents

1 Big picture **3**

2 Model environment and key objects **4**

2.1 Players and timing 4
2.2 Aggregate states 4
2.3 Investors and special depositors 5
2.4 Banks, leverage, and socially valuable lending 5
2.5 Government policy instruments 5
2.6 Public liquidity, bailouts, and insurance 6
2.7 Extension: imperfectly correlated liquidity shocks 6

3 Models and theoretical specifications (all equations + interpretation) **7**

3.1 Bank continuation 7
3.2 Bank utility 7
3.3 Social welfare 7
3.4 Pure shadow banking 8
3.5 Shadow banking with LOLR 8
3.6 Shadow banking with deposit insurance 9
3.7 Regulated banking without public insurance 9
3.8 Regulated banking with LOLR 9
3.9 Regulated banking with deposit insurance 10
3.10 Complementarity between regulation and public insurance 10
3.11 The traditional banking quadrilogy 10
3.12 Coexistence of regulated and shadow banking 11
3.13 Imperfectly correlated shocks and liquidity sharing 11
3.14 Ring-fencing 12

3.15 Central counterparties 12

4 Mechanism and policy logic (what makes the theory work) 12

4.1 What standard views miss 12

4.2 Why public liquidity is special 12

4.3 Why regulation complements LOLR 13

4.4 Why regulation complements deposit insurance 13

4.5 Why SME lending matters 13

4.6 Why shadow banking constrains policy 13

4.7 Why ring-fencing is useful 14

4.8 Why CCPs are useful 14

5 Figures and propositions 14

5.1 Figure 1: Timing for bank operations 14

5.2 Proposition 1: Optimum in the absence of regulation 15

5.3 Proposition 2: Optimum given regulation 15

5.4 Corollary 1: Complementarity 15

5.5 Proposition 3: Traditional banking system 16

5.6 Corollary 2 and Proposition 4: Coexistence and menus 16

5.7 Figure 2 and Proposition 5: Bilateral hedging 16

5.8 Proposition 6: Ring-fencing 17

5.9 Proposition 7: CCPs 17

5.10 Relationship to the literature 17

6 Short summary 17

7 Conclusion 18

7.1 Core conclusion 18

7.2 Economic intuition 18

7.3 Takeaway 18

1 Big picture

- **Question.** Why do the classic features of traditional banking tend to come together? In particular, why are retail banks typically prudentially supervised, allowed to take insured deposits, given access to lender-of-last-resort support, and involved in lending to socially important borrowers such as small and medium enterprises?
- **Setting.** The paper builds a theory of banking regulation with two sectors:
  - a **regulated retail banking sector**, where the government can monitor bank balance sheets and restrict leverage;
  - a **shadow banking sector**, where banks are not monitored and can migrate if regulation is unattractive.
- **Core challenge.** Public support to banks is useful because the government can provide liquidity when private markets cannot. But the expectation of public support also creates moral hazard: banks may lever up, become empty shells in bad states, and shift losses onto taxpayers.
- **Main contribution.** The paper explains the logic of a banking **quadrilogy**:
  1. SME lending or other socially important core services;
  2. insured deposit taking;
  3. access to lender of last resort;
  4. prudential supervision.

  The key result is that these features are complements. Regulation reduces the cost of public insurance, while access to public insurance makes regulation more valuable.
- **Why shadow banking matters.** Shadow banking is not simply another financial sector. It creates an outside option for banks. If regulation is too costly or public insurance is badly priced, banks can migrate to the unregulated sector. This migration constraint shapes optimal regulation.
- **Key conceptual distinction.**
  - **Bailouts** are ex post rescues that the government would like to avoid ex ante but cannot credibly rule out ex post.
  - **Public insurance services** such as LOLR and DI are ex ante arrangements that the government may want to offer, because they can support valuable banking activities and safe deposits.
- **Main theoretical results.**
  - Without regulation, LOLR and DI are expensive because banks exploit them by increasing leverage.
  - With regulation, leverage is constrained, so LOLR and DI become cheaper and more attractive.
  - The traditional banking system emerges when public insurance is valuable only if combined with supervision.
  - Shadow banking can coexist with regulated banking when some activities are too hard or too costly to supervise.

- Cross-exposures between regulated and shadow banks create new distortions: bogus liquidity, liquidity siphoning, and contagion.
- Ring-fencing and centralized clearing through CCPs can prevent these distortions.

- **Economic takeaway.** Traditional banking is not an arbitrary institutional bundle. It is an equilibrium response to a basic policy tradeoff: society wants banks to provide liquidity, safe deposits, and credit to important borrowers, but public support must be tied to supervision to limit moral hazard.

2 Model environment and key objects

2.1 Players and timing

- The model has two dates, $t = 0$ and $t = 1$.
- There are three groups of players:
  1. investors and depositors;
  2. bankers;
  3. the government or regulator.
- At date 0, banks choose leverage and decide whether to enter the regulated or shadow banking sector.
- At date 1, uncertainty is realized, debt is repaid subject to limited liability, and banks either continue and finance their projects or fail to continue.

2.2 Aggregate states

- The aggregate state is
$$\omega = (\omega_1, \omega_2).$$
- $\omega_1 \in \{G, B\}$ is the **fiscal state**:
  - G: good fiscal state;
  - B: bad fiscal state.
- The shadow cost of public funds is $1 + \lambda_{\omega_1}$, with
$$0 \leq \lambda_G < \lambda_B.$$
- $\omega_2 \in \{g, b\}$ is the baseline **bank liquidity state**:
  - g: banks have high revenue;
  - b: banks have low revenue.
- Bank revenue is normalized as
$$r_g = 2, \qquad r_b = 0.$$

Interpretation. The government is more reluctant to use taxpayer funds in the bad fiscal state. Banks are privately liquid in state g and illiquid in state b. The model separates the government fiscal condition from bank liquidity.

2.3 Investors and special depositors

- There are two types of investors.
- **Ordinary investors** are risk neutral. They hold uninsured claims.
- **Special depositors** demand absolutely safe claims. They are willing to pay $1 + \theta$ at date 0 for the certainty of receiving 1 at date 1.
- The mass of special depositors is $\mu \leq 1$.
- Private banks cannot create perfectly safe deposits in aggregate bad states by themselves; they need government backing.

Interpretation. Special depositors capture households or institutions that highly value safety and liquidity. Deposit insurance is valuable because it allows banks to produce a safe asset that the private sector cannot fully guarantee.

2.4 Banks, leverage, and socially valuable lending

- Each banker has a project that requires one unit of investment at date 1.
- The project represents lending to SMEs or another core financial service.
- The banker is indispensable for the project.
- Continuation gives the banker private benefit b and creates social value β for bank stakeholders, interpreted as borrowers or firms that depend on credit.
- At date 0, bank i issues unsecured wholesale debt with face value $d^i_{\omega_1}$, contingent on the verifiable fiscal state.
- Debt repayment at date 1 is limited by bank revenue:

$$\min\{d^i_{\omega_1}, r_{\omega_2}\}.$$

Interpretation. Leverage is the key bank moral-hazard margin. A bank that expects to be rescued has an incentive to issue more debt and leave fewer resources available for depositors or reinvestment.

2.5 Government policy instruments

- The government can offer a regulatory contract

$$\{\tau_0, k, l, m, \{d_{\omega_1}\}\},$$

  where:

  - τ_0 is a date-0 transfer or fee;

- $k = 1$ means deposit insurance is provided;
- $l = 1$ means lender-of-last-resort support is provided;
- $m = 1$ means the bank is monitored or regulated;
- d_{ω_1} is the prescribed leverage level.

- Regulated banks face leverage constraints.
- Shadow banks are not monitored and freely choose leverage.
- Pure shadow banking is always available as an outside option.

Interpretation. The government cannot simply impose regulation without regard to migration. Banks must prefer the regulated contract to their shadow-banking outside option.

2.6 Public liquidity, bailouts, and insurance

- **Bailouts** occur when the government has not committed ex ante to support the bank, but still rescues it ex post because failure would be socially costly.
- **LOLR** is an ex ante commitment to help a bank continue when it lacks liquidity.
- **DI** is an ex ante commitment to make special depositors whole.
- Public transfers are costly because they require distortionary public funds.

The expected deadweight loss from one unit of public transfer in bank liquidity state ω_2 is

$$\Delta_{\omega_2} \equiv p_{\omega_2} \sum_{\omega_1 \in \{G,B\}} p_{\omega_1|\omega_2} \lambda_{\omega_1}.$$

The average shadow cost is

$$\bar{\lambda} \equiv \Delta_g + \Delta_b = p_G \lambda_G + p_B \lambda_B.$$

Interpretation. The cost of public insurance depends both on how often the government must pay and on whether those payments happen in a good or bad fiscal state.

2.7 Extension: imperfectly correlated liquidity shocks

- In the baseline model, all bank liquidity shocks are perfectly correlated.
- Later, the paper introduces intermediate liquidity states m_H and m_T.
- In these states, one type of bank has cash and the other does not.
- This creates a potential role for liquidity sharing through swaps, credit lines, guarantees, and other cross-exposures.

Interpretation. The extension introduces the reason why banks form cross-exposures. It also introduces the danger that those exposures are used to game regulation rather than to provide genuine insurance.

3 Models and theoretical specifications (all equations + interpretation)

3.1 Bank continuation

Let $x_\omega^i \in \{0,1\}$ denote whether bank i continues and finances its date-1 project in state ω.

- $x_\omega^i = 1$ means the bank continues.
- $x_\omega^i = 0$ means the bank does not continue.
- Continuation requires one unit of investment.

A bank continues if it has enough private cash or receives public liquidity support. In the baseline, public support may come from:

- bailout in fiscal state G;
- LOLR if $l = 1$;
- deposit-insurance transfers if special deposits are insured.

Interpretation. The government's inability to commit not to rescue in the good fiscal state creates an implicit bailout option. This implicit option matters even before formal LOLR or DI is introduced.

3.2 Bank utility

Bank i's utility is

$$U_i \equiv \mathbb{E}_\omega \left[b x_\omega^i + \min\{d_{\omega_1}^i, r_{\omega_2}\} + (1+\theta)\mu_i + \tau_0^i \right].$$

- $b x_\omega^i$ is the private benefit of continuation.
- $\min\{d_{\omega_1}^i, r_{\omega_2}\}$ is the date-0 value of unsecured debt issuance.
- $(1+\theta)\mu_i$ is the revenue from issuing insured safe deposits.
- τ_0^i is the transfer from the government.

Interpretation. Banks like continuation, but they also like to raise debt and deposit financing. If public support protects continuation even when the bank is highly levered, the bank has an incentive to maximize debt issuance.

3.3 Social welfare

Total welfare is

$$W \equiv \mathbb{E}_\omega \left[\int_0^1 \Big((b+\beta-1)x_\omega^i + r_{\omega_2} + \theta\mu_i - cm_i \right.$$
$$\left. - \lambda_{\omega_1} \max\{0, x_\omega^i + \mu_i + \min\{d_{\omega_1}^i, r_{\omega_2}\} - r_{\omega_2}\} \Big) \mathrm{d}i \right].$$

- $(b+\beta-1)x^i_\omega$ is net surplus from continuation.
- r_{ω_2} is bank revenue.
- $\theta\mu_i$ is the surplus from safe deposits.
- cm_i is the cost of supervision.
- The last term is the deadweight cost of public funds.

Interpretation. The planner values bank continuation and safe deposits, but dislikes costly supervision and costly public transfers. The central question is when supervision is worth its cost because it reduces the fiscal cost of insurance and bailouts.

3.4 Pure shadow banking

Under pure shadow banking, banks are not regulated and do not receive formal LOLR or DI.

The bank chooses high leverage in fiscal state G:

$$d_G = 2,$$

so that it extracts the full high-state revenue while expecting bailout support if it lacks cash for continuation.

In fiscal state B, without LOLR, the bank chooses

$$d_B = 1,$$

so that it can still continue when it has revenue $r_g = 2$.

The bank's utility is

$$U^{\text{SB}} = (1 - p_{Bb})b + [2p_{Gg} + p_{Bg}].$$

Social welfare is

$$W^{\text{SB}} = (1 - p_{Bb})(b + \beta - 1) + 2p_g - \lambda_G p_G.$$

Interpretation. Pure shadow banks exploit expected bailouts in the good fiscal state but cannot count on public support in the bad fiscal state. They continue except in state (B, b).

3.5 Shadow banking with LOLR

If shadow banks receive LOLR, they set

$$d_G = d_B = 2.$$

The net social benefit of LOLR in the shadow sector is

$$W^{\text{SBL}} - W^{\text{SB}} = p_{Bb}(b + \beta - 1) - p_B\lambda_B.$$

Thus LOLR is optimal without regulation if and only if

$$W^{\text{SBL}} > W^{\text{SB}} \iff b + \beta - 1 > \lambda_B \frac{p_B}{p_{Bb}}.$$

Interpretation. LOLR helps by allowing continuation in state (B, b), but it is costly because unregulated banks use it to become empty shells in the bad fiscal state. This is the moral-hazard cost of public liquidity.

3.6 Shadow banking with deposit insurance

With DI, the net social benefit per unit of special deposits is the depositor surplus θ minus the average fiscal cost $\bar{\lambda}$.

The net benefit is

$$W^{\mathrm{SBD}} - W^{\mathrm{SB}} = \mu(\theta - \bar{\lambda}).$$

Thus DI is optimal without regulation if and only if

$$W^{\mathrm{SBD}} > W^{\mathrm{SB}} \quad \Longleftrightarrow \quad \theta > \bar{\lambda}.$$

Interpretation. In an unregulated bank, deposit insurance is expensive because the bank can pledge or divert its cash through leverage. The government must pay even when the bank has revenue, so the relevant fiscal cost is the average public-funds cost.

3.7 Regulated banking without public insurance

Regulation constrains leverage. In the absence of LOLR and DI, the regulated bank is constrained to choose

$$d_G = d_B = 1.$$

The welfare gain from regulation relative to pure shadow banking is

$$W^{\mathrm{RB}} - W^{\mathrm{SB}} = p_{Gg}\lambda_G - c.$$

Interpretation. Regulation prevents unnecessary bailouts in state (G, g) by stopping the bank from extracting all cash through debt. But supervision costs c, so regulation is worthwhile only if avoided bailout costs are large enough.

3.8 Regulated banking with LOLR

With regulation, LOLR no longer induces the same excessive leverage. The bank is constrained to maintain enough resources in the good liquidity state.

The net benefit of LOLR under regulation is

$$W^{\mathrm{RBL}} - W^{\mathrm{RB}} = [W^{\mathrm{SBL}} - W^{\mathrm{SB}}] + p_{Bg}\lambda_B = p_{Bb}[b + \beta - (1 + \lambda_B)].$$

Thus LOLR is optimal with regulation if and only if

$$W^{\mathrm{RBL}} > W^{\mathrm{RB}} \quad \Longleftrightarrow \quad b + \beta > 1 + \lambda_B.$$

Interpretation. Regulation removes the wasteful use of LOLR in state (B, g), where the bank has cash. With supervision, LOLR is used only when it is genuinely needed.

3.9 Regulated banking with deposit insurance

With DI and regulation, leverage is constrained so that the bank can repay depositors when it has revenue. One feasible constraint is

$$d_G = d_B = 1 - \mu.$$

The net benefit of deposit insurance under regulation is

$$W^{\text{RBD}} - W^{\text{RB}} = [W^{\text{SBD}} - W^{\text{SB}}] + \mu\Delta_g = \mu(\theta - \Delta_b).$$

Thus DI is optimal with regulation if and only if

$$W^{\text{RBD}} > W^{\text{RB}} \quad \Longleftrightarrow \quad \theta > \Delta_b.$$

Interpretation. Regulated banks pay depositors out of their own resources in the good liquidity state. Deposit insurance is called only in the bad liquidity state. Therefore the relevant public cost is Δ_b, not $\bar{\lambda}$.

3.10 Complementarity between regulation and public insurance

The planner chooses $k, l, m \in \{0, 1\}$ to maximize

$$\max_{k,l,m} W(k, l, m),$$

where

$$\begin{aligned} W(k, l, m) = {} & W^{\text{SB}} + [W^{\text{SBL}} - W^{\text{SB}}]l + [W^{\text{SBD}} - W^{\text{SB}}]k \\ & + [W^{\text{RB}} - W^{\text{SB}}]m \\ & + \big([W^{\text{RBL}} - W^{\text{RB}}] - [W^{\text{SBL}} - W^{\text{SB}}]\big)ml \\ & + \big([W^{\text{RBD}} - W^{\text{RB}}] - [W^{\text{SBD}} - W^{\text{SB}}]\big)mk. \end{aligned}$$

The interaction terms are positive:

$$W^{\text{RBL}} - W^{\text{RB}} = W^{\text{SBL}} - W^{\text{SB}} + p_{Bg}\lambda_B > W^{\text{SBL}} - W^{\text{SB}},$$

and

$$W^{\text{RBD}} - W^{\text{RB}} = W^{\text{SBD}} - W^{\text{SB}} + \mu\Delta_g > W^{\text{SBD}} - W^{\text{SB}}.$$

Interpretation. Regulation and public insurance are complements. Regulation makes LOLR and DI cheaper by reducing leverage-based moral hazard. Conversely, the presence of LOLR and DI increases the value of supervision.

3.11 The traditional banking quadrilogy

A traditional banking system with regulation, LOLR, and DI is optimal, while those public insurance services would not be granted to shadow banks, under conditions such as

$$0 < p_{Bb}[b + \beta - (1 + \lambda_B)] < p_{Bg}\lambda_B,$$

$$\Delta_b < \theta < \bar{\lambda} = \Delta_g + \Delta_b,$$

and

$$c \leq c^* = p_{Gg}\lambda_G + p_{Bb}[b + \beta - (1 + \lambda_B)] + \mu(\theta - \Delta_b).$$

Interpretation.

- The first condition says LOLR is socially valuable when supervised, but too costly when unsupervised.
- The second condition says DI is valuable when supervised, but too costly when unsupervised.
- The third condition says supervision is not too costly relative to its three benefits: avoided bailouts, cheaper LOLR, and cheaper DI.

This is the formal basis for the four-pillar system.

3.12 Coexistence of regulated and shadow banking

If some activities have low monitoring cost and others have infinite or very high monitoring cost, the economy can have both sectors:

- low-monitoring-cost banks enter the regulated sector and receive LOLR and DI;
- high-monitoring-cost banks remain in shadow banking and receive no public liquidity support;
- special depositors are served by regulated banks.

Interpretation. Shadow banking can arise because some activities are too complex to supervise. The model therefore allows shadow banking to be more than pure regulatory arbitrage, even though arbitrage remains a central constraint.

3.13 Imperfectly correlated shocks and liquidity sharing

The paper then expands the liquidity state to

$$\omega_2 \in \{g, b, m_H, m_T\}.$$

There are two types of banks, H and T.

- In state m_H, type H banks receive revenue 2 and type T banks receive 0.
- In state m_T, type T banks receive revenue 2 and type H banks receive 0.

Optimal liquidity sharing sets

$$d_{\omega_1 m_H} = d_{\omega_1 m_T} = 0 \qquad \text{for } \omega_1 \in \{G, B\},$$

and matches each bank with a natural counterparty of the opposite type.

Interpretation. If an H bank and a T bank insure each other, the cash-rich bank can transfer one unit to the cash-poor bank in the intermediate liquidity state. Both banks can then continue without public funds.

3.14 Ring-fencing

Ring-fencing means that a regulated bank cannot be exposed to the default of a shadow bank. In the model, it prevents regulated banks from signing liquidity-sharing contracts with unregulated counterparties.

Interpretation. Without ring-fencing, a regulated bank can satisfy the appearance of having a liquidity contract while actually choosing a correlated shadow-bank counterparty that fails exactly when it is needed. This creates bogus liquidity and siphons liquidity from the regulated sector to the shadow sector.

3.15 Central counterparties

If the regulator cannot observe correlations even within the regulated sector, bilateral contracts are still vulnerable. Regulated banks can choose correlated regulated counterparties and create bogus liquidity.

A CCP solves this by centralizing liquidity sharing:

- regulated banks must participate in the CCP;
- the CCP pools contributions and reallocates liquidity to banks that need it;
- shadow banks are excluded by ring-fencing;
- individual banks cannot choose correlated counterparties to game the system.

Interpretation. Ring-fencing prevents regulated-to-shadow gaming. A CCP prevents within-regulated-sector gaming when correlation is hard to observe.

4 Mechanism and policy logic (what makes the theory work)

4.1 What standard views miss

- A simple view says regulation is costly and shadow banking is useful because it avoids regulatory burdens.
- Another simple view says shadow banking is bad because it evades regulation and creates systemic risk.
- This paper combines both ideas. Shadow banking can be useful when activities are genuinely hard to supervise, but it also creates a migration constraint and opportunities for regulatory arbitrage.

4.2 Why public liquidity is special

- Private investors cannot commit to provide future funds in bad aggregate states.
- The government can raise taxes and therefore has a unique ability to create liquidity.
- This ability makes LOLR and DI valuable.

- But public liquidity is costly because taxpayer funds have a shadow cost, especially in bad fiscal states.

Mechanism. Public liquidity support is valuable because it supports socially important bank activities, but it creates a taxpayer put option that banks can exploit.

4.3 Why regulation complements LOLR

- Without regulation, LOLR causes banks to increase leverage in fiscal state B.
- The bank then has less cash available when liquidity is needed.
- The government must provide more support.
- Regulation restricts leverage, so LOLR is called only in genuinely bad liquidity states.

Mechanism. Regulation converts LOLR from a subsidy to excessive leverage into targeted liquidity insurance.

4.4 Why regulation complements deposit insurance

- Special depositors value safety.
- Deposit insurance creates safe deposits.
- Without regulation, banks can lever up and leave the government responsible for repayment even when the bank had cash earlier.
- With regulation, the bank preserves enough resources to repay depositors in good liquidity states.

Mechanism. Regulation lets the government piggyback on bank balance sheets. The government provides insurance only in states where bank cash is truly insufficient.

4.5 Why SME lending matters

- The government values continuation because SMEs or other core borrowers depend on bank credit.
- This makes ex post bailouts tempting.
- Banks anticipate this and choose leverage accordingly.

Mechanism. The social value of lending creates the time-inconsistency problem. The government wants to support credit ex post, but that support distorts bank leverage ex ante.

4.6 Why shadow banking constrains policy

- Banks can choose the shadow sector if regulated banking is unattractive.
- Public insurance may need to be priced attractively enough to keep banks regulated.

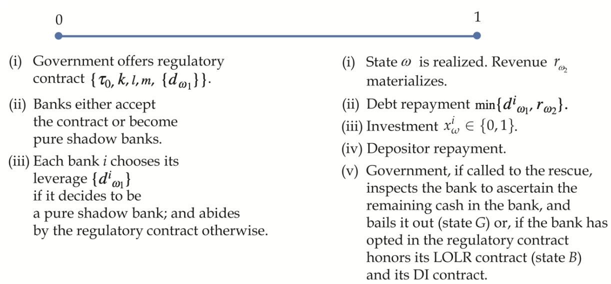

FIGURE 1
Timing for bank's operations.

- But public insurance cannot be offered freely to shadow banks, because unregulated banks would exploit it.

Mechanism. The regulated contract must solve two problems at once: control moral hazard and satisfy bank participation relative to shadow banking.

4.7 Why ring-fencing is useful

- Cross-exposures can be beneficial when they transfer liquidity from cash-rich to cash-poor banks.
- But cross-exposures can also be fake insurance.
- A regulated bank can contract with a correlated shadow bank that defaults exactly when liquidity is needed.
- This raises the expected cost of public support.

Mechanism. Ring-fencing blocks contracts that make regulated banks dependent on unregulated counterparties.

4.8 Why CCPs are useful

- If the regulator can observe correlation among regulated banks, bilateral contracts can implement optimal risk sharing.
- If the regulator cannot observe correlation, bilateral contracts are gameable.
- A CCP removes the choice of counterparty and centralizes risk sharing.

Mechanism. A CCP prevents banks from choosing the wrong counterparty on purpose.

5 Figures and propositions

5.1 Figure 1: Timing for bank operations

Read:

- The government first offers a regulatory contract.

- Banks then choose whether to be regulated or shadow banks.
- Banks choose leverage.
- At date 1, the fiscal and liquidity states are realized.
- Banks repay debt subject to limited liability and either continue or fail to continue.

Takeaway. The timing makes leverage an ex ante moral-hazard choice. Banks choose leverage before knowing whether they will need public support.

5.2 Proposition 1: Optimum in the absence of regulation

Read:

- Without regulation, LOLR is optimal only if the continuation benefit is large enough to offset the fiscal cost created by unregulated leverage.
- Without regulation, DI is optimal only if special depositors' valuation of safety exceeds the average public cost of providing it.
- There is no direct complementarity between LOLR and DI in the unregulated sector.

Takeaway. Public insurance is hard to justify in shadow banking because there is no supervisory control over balance-sheet risk.

5.3 Proposition 2: Optimum given regulation

Read:

- With regulation, LOLR is optimal under the weaker condition $b + \beta > 1 + \lambda_B$.
- With regulation, DI is optimal under the weaker condition $\theta > \Delta_b$.
- Regulation improves the terms on which public liquidity can be offered.

Takeaway. Regulation does not just reduce bailouts. It makes public insurance services more efficient.

5.4 Corollary 1: Complementarity

Read:

- The net benefit of LOLR is higher when the bank is regulated.
- The net benefit of DI is higher when the bank is regulated.
- The positive interaction terms in $W(k, l, m)$ are the mathematical expression of complementarity.

Takeaway. The four pillars of traditional banking are linked by economies of scope in supervision.

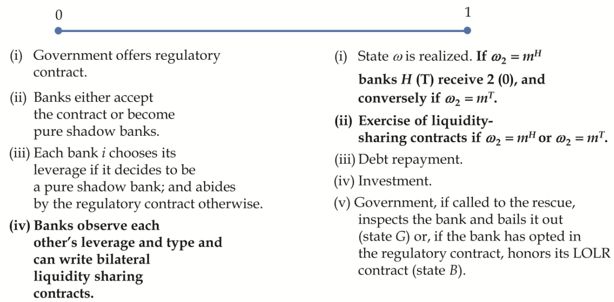

Figure 2
Timing under bilateral hedging.

5.5 Proposition 3: Traditional banking system

Read:

- A traditional banking system emerges when LOLR and DI are socially valuable only under regulation.
- The supervision cost must be below a threshold c^*.
- The threshold includes three benefits: avoided bailouts, cheaper LOLR, and cheaper DI.

Takeaway. The model rationalizes why the government ties access to deposit insurance and lender-of-last-resort support to prudential supervision.

5.6 Corollary 2 and Proposition 4: Coexistence and menus

Read:

- If some banks or activities are costly to monitor, regulated and shadow banking can coexist.
- If monitoring is valuable even without special deposits, one-size-fits-all regulation is enough.
- If monitoring is valuable mainly because of special deposits, the optimal system may involve a menu: regulated deposit-taking banks and shadow non-deposit-taking banks.

Takeaway. Shadow banking is not eliminated in the optimum when some activities are too hard to supervise or when it is efficient to separate deposit-taking from other activities.

5.7 Figure 2 and Proposition 5: Bilateral hedging

Read:

- The extended model adds liquidity-sharing contracts at date 0.
- Natural counterparties are banks with opposite liquidity exposures.
- Optimal risk sharing requires natural counterparties and zero leverage in intermediate liquidity states.

Takeaway. Liquidity sharing is socially valuable when it reduces public transfers and allows both banks to continue.

5.8 Proposition 6: Ring-fencing

Read:

- Without ring-fencing, regulated banks have incentives to contract with correlated shadow banks.
- These contracts create bogus liquidity: apparent insurance that fails exactly when needed.
- They also create liquidity siphoning from the regulated sector to the shadow sector.
- If ring-fencing is imposed and the regulator can monitor correlations, bilateral contracts can implement the optimum.

Takeaway. Ring-fencing is needed to protect the regulated sector from hidden exposure to shadow-bank default.

5.9 Proposition 7: CCPs

Read:

- Ring-fencing alone is not enough if the regulator cannot observe correlations even among regulated banks.
- Bilateral contracts can still be gamed by matching with correlated counterparties.
- Mandating clearing through a CCP implements optimal liquidity sharing by centralizing the matching and transfer system.

Takeaway. CCPs are useful because they prevent banks from selecting counterparties in a way that maximizes the taxpayer put.

5.10 Relationship to the literature

Read:

- The paper connects to regulatory-arbitrage theories of shadow banking.
- It also relates to theories of safe-asset demand and comparative advantage in shadow banking.
- Its distinctive contribution is to explain the complementarity among regulation, LOLR, DI, and core banking services, and to formalize ring-fencing and CCPs as structural remedies.

Takeaway. The paper does not treat traditional banking institutions as exogenous. It derives them as jointly optimal responses to public-liquidity provision and shadow-banking migration.

6 Short summary

- **Method.** Build a mechanism-design model in which banks choose leverage, can migrate to shadow banking, and may receive public liquidity support.
- **Core friction.** The government values bank continuation and safe deposits, but public funds are costly and banks exploit expected support by increasing leverage.

- **Main mechanism.** Supervision reduces leverage-based moral hazard, so it lowers the cost of LOLR and DI.
- **Traditional banking result.** The four pillars of traditional banking emerge together: SME lending, insured deposits, LOLR, and prudential supervision.
- **Shadow banking result.** Shadow banking acts as a migration constraint and may coexist with regulated banking when some activities are difficult to monitor.
- **Structural remedies.** When banks share liquidity through cross-exposures, ring-fencing and CCPs prevent bogus liquidity, liquidity siphoning, and contagion.
- **Overall lesson.** Public insurance should not be separated from supervision. The same logic that justifies deposit insurance and lender-of-last-resort support also justifies prudential oversight and structural limits on cross-sector exposures.

7 Conclusion

7.1 Core conclusion

The paper shows that the traditional banking system is a coherent institutional package. The government can create liquidity in bad times, but this ability is costly and creates moral hazard. Prudential supervision lowers the fiscal cost of public liquidity services. Therefore, LOLR, deposit insurance, regulation, and core lending naturally belong together.

7.2 Economic intuition

A bank that expects public support will choose too much leverage unless constrained. This makes public insurance expensive. Regulation limits leverage and lets public liquidity be used only when bank liquidity is truly insufficient. Shadow banking weakens this system because banks can migrate away from regulation or use shadow counterparties to create fake insurance. Ring-fencing and CCPs address these additional distortions by controlling the structure of interbank exposures.

7.3 Takeaway

The paper turns the four pillars of traditional banking into a formal theory. Its message is that the boundary between traditional and shadow banking should be designed around the complementarity between supervision and public insurance. In the extended model, this becomes a broader **hexalogy**: core banking services, insured deposits, prudential supervision, LOLR, ring-fencing, and centralized clearing all work together to control the taxpayer put created by financial intermediation.